

# Grocery List Manager

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## 1. Introduction

A **Grocery List Manager** is a web-based application designed to help families or groups collaboratively manage their shopping lists in real time.

In many households, grocery planning is often unorganized, leading to duplicate purchases or missing essential items. Without a shared system, communication gaps occur between family members.

This platform provides a **real-time collaborative solution** where multiple users can manage a shared grocery list, ensuring better coordination and efficient shopping.

The project demonstrates concepts such as **real-time synchronization, collaborative applications, CRUD operations, and full-stack web development**.

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## 2. Description

The Grocery List Manager allows users to **create and manage shared grocery lists** with real-time updates across multiple devices.

The application enables users to:

- Create shared lists for families or groups
- Add, edit, and delete grocery items
- Categorize items (Vegetables, Dairy, Snacks, etc.)
- Mark items as purchased
- View updates in real time across all devices

Each group has:

- A private shared list
- Access control for members only

The platform ensures **better organization, avoids duplication, and improves shopping efficiency.**

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### 3. Problem Statement

Families often face issues while managing grocery shopping due to:

- Lack of a shared list
- Duplicate purchases
- Missing important items
- Poor coordination among members

The challenge is to build a system that:

- Allows collaborative list management
- Supports real-time updates across devices
- Enables categorization of items
- Allows marking items as purchased
- Ensures privacy for each group

The solution should provide a **simple, real-time, and collaborative grocery management system.**

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### 4. Scenario

Consider a family preparing for weekly grocery shopping.

- One member adds items like milk, bread, and fruits to the list.
- Another member updates the list by adding snacks.
- At the store, a member marks items as purchased.

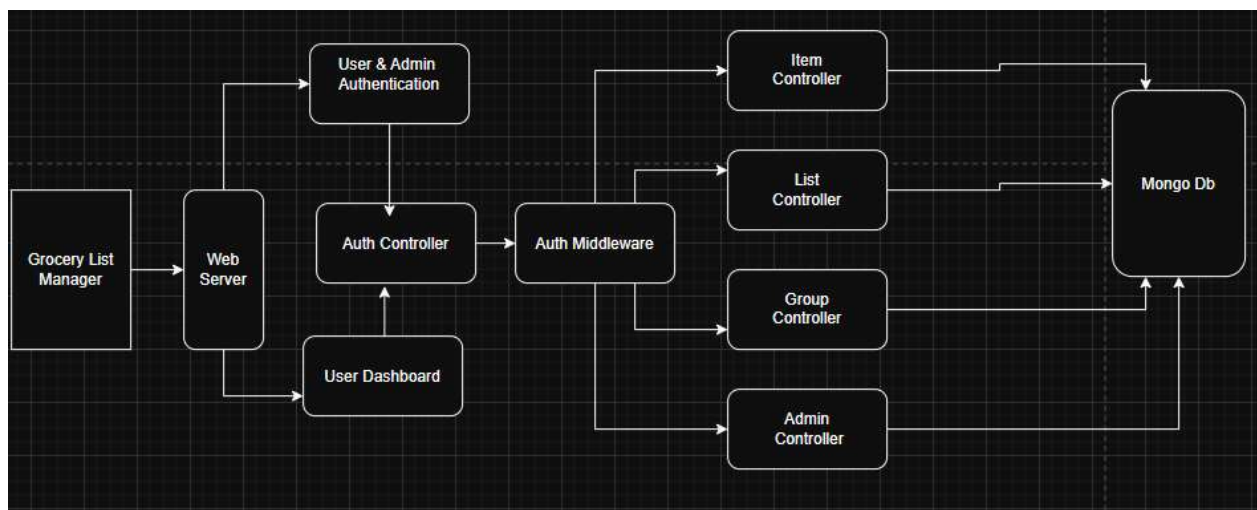
The system:

- Instantly updates the list for all members
- Shows purchased items as crossed out

This ensures:

- No duplicate purchases
- No missed items
- Smooth coordination among family members

## 5. Architecture



The platform follows a **three-tier architecture with real-time synchronization**.

## Frontend Layer

- Shared grocery list UI grouped by categories
- Add/edit/delete item interface
- Checkbox for marking items as purchased
- Real-time updates (auto-refresh without reload)
- Developed using **React.js**

## Backend Layer

- Handles list and item management
- Manages group-based access control
- Provides APIs for CRUD operations
- Implements real-time sync using **SSE / polling**
- Developed using **Node.js + Express.js**

## Database Layer

- Stores users, groups, lists, and items
- Implemented using **MongoDB**

## Communication Flow

Frontend (React)

↓ API Requests / SSE

Backend (Node.js + Express)

↓ Database Queries

MongoDB Database

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## 6. Project Flow (System Flow)

### Step 1: User Registration/Login

Users create accounts and join a group.

### Step 2: List Creation

A shared grocery list is created for the group.

### Step 3: Item Management

Users add, edit, or delete items.

### Step 4: Categorization

Items are grouped by category.

### Step 5: Real-time Sync

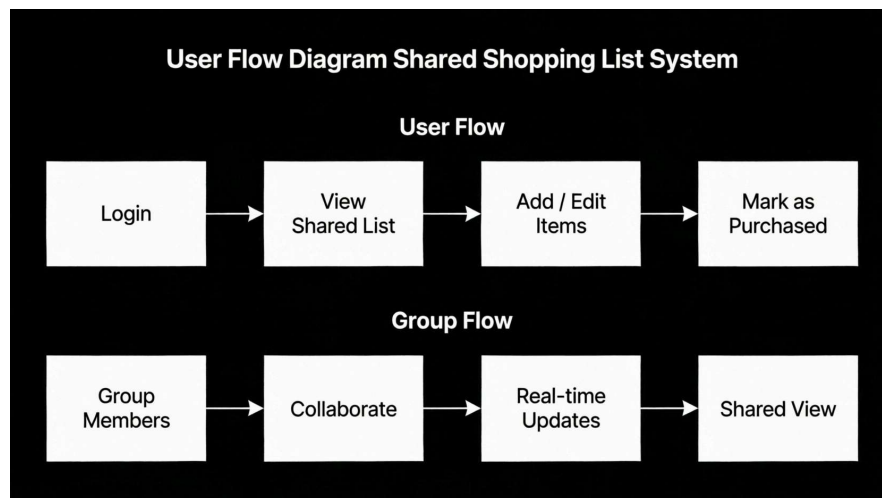
Updates are synced across all devices.

### Step 6: Purchase Tracking

Users mark items as purchased.

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## 7. User Flow



### User Flow

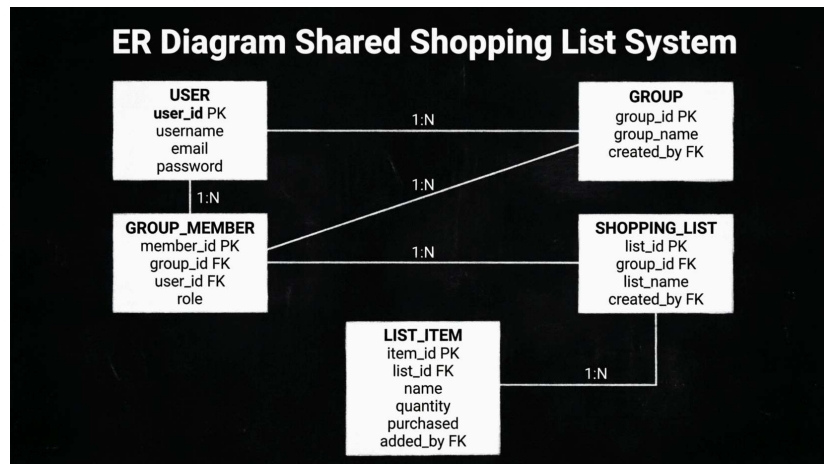
User → Login → View Shared List → Add/Edit Items → Mark as Purchased

### Group Flow

Group Members → Collaborate → Real-time Updates → Shared View

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## 8. ER Diagram



Entities:

- User - stores user account information and authentication details
- Group - stores shopping group details for collaboration
- Group\_Member - junction table tracking which users belong to which groups with roles
- Shopping\_List - stores shared shopping lists within groups
- List\_Item - stores individual items with quantity, purchased status, and tracking

Relationships:

- User → Group\_Member (1:N) - One user can be a member of many groups
- Group → Group\_Member (1:N) - One group can have many members
- Group → Shopping\_List (1:N) - One group can have multiple shopping lists
- Shopping\_List → List\_Item (1:N) - One shopping list contains many items
- User → Shopping\_List (1:N) - One user can create many shopping lists

## 9. Pre-Requisites

Frontend (React)

- React Components

- React Hooks
- State management
- Axios

### **Backend (Node.js & Express)**

- REST APIs
- CRUD operations
- Middleware
- Real-time concepts (SSE / polling)

### **Database (MongoDB)**

- Schema design
- CRUD operations
- Mongoose ODM

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## **10. Required Technologies**

### **Frontend**

- React.js
- TailwindCSS / Material UI
- Axios

### **Backend**

- Node.js

- Express.js

## Database

- MongoDB
- Mongoose

## Real-time

- Server-Sent Events (SSE) / Polling

## Tools

- Git
- GitHub
- Postman

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# 11. Suggested Database Collections

## Users

- `_id`
- `name`
- `email`
- `password`

## Groups

- `_id`

- groupName
- members (userIds)

## Lists

- \_id
- groupId
- createdAt

## Items

- \_id
- listId
- name
- category
- isPurchased
- updatedAt

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# 12. Key Features

## Collaborative Lists

- Shared grocery lists
- Multi-user access

## Item Management

- Add/edit/delete items

- Categorization

### **Real-time Sync**

- Instant updates across devices

### **Purchase Tracking**

- Mark items as purchased
- Visual indicators (cross-out)

### **Privacy**

- Group-based access control
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## **13. Optional Advanced Features**

- Push notifications for updates
  - Voice-based item addition
  - Smart suggestions (AI-based)
  - Offline support with sync
  - Barcode scanner for products
  - Budget tracking
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## **14. Learning Outcomes**

By completing this project, developers will learn:

- Full-stack MERN development

- Real-time data synchronization
- Collaborative application design
- CRUD operations and APIs
- Access control and group management
- UI/UX for shared systems
- Database schema design